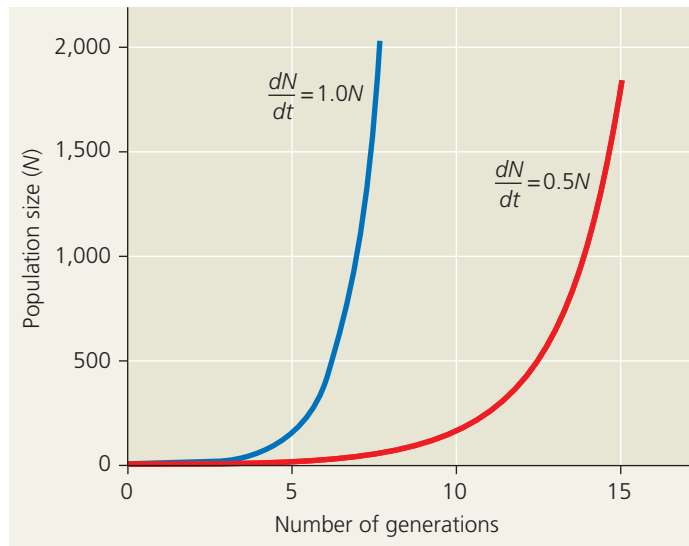


▼ **Figure 53.7 Population growth predicted by the exponential model.** This graph compares growth in a population with $r = 1.0$ (blue curve) to growth in a population with $r = 0.5$ (red curve).



? How many generations does it take these populations to reach a size of 1,500 individuals?

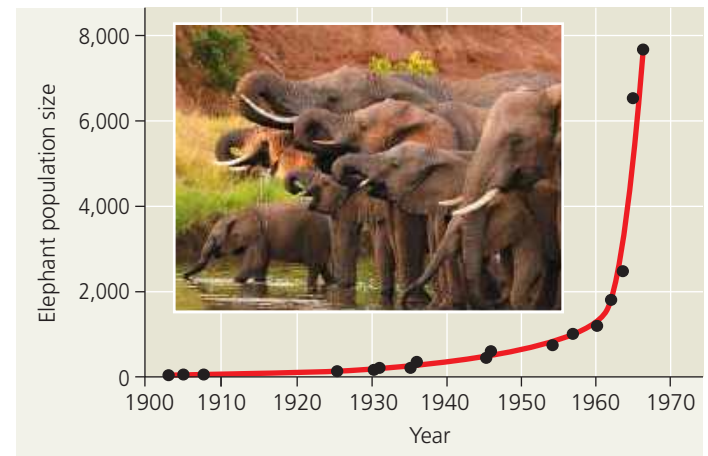
➔ Mastering Biology BioFlix® Animation: Exponential Growth

In this equation, dN/dt represents the rate at which the population is increasing in size at each moment in time, akin to how a glance at the speedometer of a car reveals the speed at that instant in time. As seen in the equation, dN/dt equals the current population size, N , multiplied by a constant, r . Ecologists refer to r as the **intrinsic rate of increase**, the per capita rate at which an exponentially growing population increases in size at each instant in time.

The size of a population that is growing exponentially increases at a constant rate per individual, resulting eventually in a J-shaped growth curve when population size is plotted over time (Figure 53.7). Although the per capita rate of population growth is constant (and equals r), more new individuals are added per unit of time when the population is large than when it is small; thus, the curves in Figure 53.7 get progressively steeper over time. This occurs because population growth depends on N as well as r , and hence more individuals are added to larger populations than to small ones growing at the same per capita rate. It is also clear from Figure 53.7 that a population with a higher intrinsic rate of increase ($dN/dt = 1.0N$) will grow faster than one with a lower intrinsic rate of increase ($dN/dt = 0.5N$).

The J-shaped curve of exponential growth is characteristic of some populations that are introduced into a new environment or whose numbers have been drastically reduced by a catastrophic event and are rebounding. For example, the population of elephants in Kruger National Park, South Africa, grew exponentially for approximately 60 years after they were first protected from hunting (Figure 53.8). The increasingly

▼ **Figure 53.8 Exponential growth in the African elephant population of Kruger National Park, South Africa.**



large number of elephants eventually caused enough damage to vegetation in the park that a collapse in their food supply was likely. To protect other species and the park ecosystem before that happened, park managers began limiting the elephant population by giving birth control medication to females and by exporting elephants to other countries.

CONCEPT CHECK 53.2

1. Explain why a constant per capita rate of growth (r) for a population produces a curve that is J-shaped.
2. Where is exponential growth by a plant population more likely—in an area where a forest was destroyed by fire or in a mature, undisturbed forest? Why?
3. **WHAT IF?** In 2018, the United States had a population of 327 million people. If the (annual) per capita change in population size ($r_{\Delta t}$) was 0.005, how many people were added to the population that year (ignoring immigration and emigration)? What would you need to know to determine whether the United States is currently experiencing exponential growth?

For suggested answers, see Appendix A.

CONCEPT 53.3

The logistic model describes how a population grows more slowly as it nears its carrying capacity

The exponential growth model assumes that resources remain abundant, which is rarely the case in the real world. Instead, as the size of a population increases, each individual has access to fewer resources. Ultimately, there is a limit to the number of individuals that can occupy a habitat. Ecologists define the **carrying capacity**, symbolized by K , as the maximum population size that a particular environment can sustain. Carrying capacity varies over space and time with the abundance of limiting resources. Energy, shelter, refuge from predators, nutrient availability, water, and